

DISCRETE MATHEMATICS (3-1-0-4)

COURSE CODE-MC123

Course placement: This is a core course for first year students of BTech MnC program.

Course format: It is 3 hours lecture every week and 1 tutorial per week.

Course content: The primary aim of this course is to introduce students to the fundamentals of Discrete mathematics. In this course we cover the following chapter:

Basic discrete structures: Set operations, Functions, Relations and their properties, Equivalence relations, Closure of relations, Partial orderings, Sequences and summations, Matrices (3 lectures).

Mathematical logic and Proof techniques: Proposition logic, predicate logic and quantifiers, logic operators, logic proposition Introduction to different standard proof techniques (3 lectures).

Combinatorics: Combinatorial Principles and Techniques, permutations and combinations, binomial coefficients and identities, Pigeon hole principle, inclusion exclusion principle, Recurrence relations, Solving linear recurrence relations and applications (8 lectures).

Number Theory: Induction, strong induction, well-ordered property, structural induction, and generalized induction Modular arithmetic, primes and representation of integers, linear congruences, cryptography, RSA cryptosystem (6 lectures)

Probability: Basic discrete probability, Probabilistic counting, conditional probability, Baye's theorem, random variables (6 lectures).

Graph Theory: Undirected graph, Representation on the computer: adjacency matrix, Handshaking lemma, Euler path and circuit, Hamiltonian cycle, Complete graph, cycles, bipartite graph, planar graphs, Euler formula, shortest path algorithm, Traveling salesman problem, trees. properties of trees, spanning trees (8 lectures).

Boolean Algebra: Boolean functions, logic gates, minimization expressions, K-maps (3 lectures).

Modeling Computation: DFA, NFA, languages, regular expressions, grammars, PDF, introduction to Turing machines. (3 lectures).

Suggested Textbook/references

- Discrete Mathematics, Norman. L. Biggs, Oxford science publication.

